0.1 Product Space

0.1.1 Finite Product Space

Definition 0.1.1. Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be Topological Spaces. Define a Topology on $X \times Y$:

$$\mathcal{T}_P \stackrel{\text{def}}{=} \{U \times V \subset X \times Y \mid U \in \mathcal{T}_X, V \in \mathcal{T}_Y\}$$

The Topological Space $(X \times Y, \mathcal{T}_P)$ is called the *Product Space*.

Lemma 0.1.2. Suppose that $(X \times Y, \mathcal{T}_p)$ is a Product Space for X and Y. Then, the Projection map

$$\pi_X : X \times Y \to X : (x, y) \mapsto x$$

is Continuous Open Onto map.

Proof. Surjection is Clear,
Continuous: For any open $\mathcal{U} \in \mathcal{T}_X$, $\pi_X^{-1}[\mathcal{U}] = \mathcal{U} \times Y \in \mathcal{T}_P$.
Open: For any open $O \times V \in \mathcal{T}_P$, $\pi_X[O \times V] = O \in \mathcal{T}_X$. Thus, it is Open map. □

Theorem 0.1.3. $\mathcal{T}_p$ is the samllest Topology such that the Projection maps are Continuous.

Proof. Suppost that $\mathcal{T}'$ is Topology on $X \times Y$ such that π_X, π_Y are Continous map.
Then, for any open $U \in \mathcal{T}_X, V \in \mathcal{T}_Y$, $U \times Y = \pi_X^{-1}[U] \in \mathcal{T}'$ and $X \times V = \pi_Y^{-1}[V] \in \mathcal{T}'$
Thus, the finite intersection $U \times Y \cap X \times V = U \times V \in \mathcal{T}'$. That is, $\beta_p \subset \mathcal{T}'$. □

Theorem 0.1.4. Let X, and $\{Y_i\}_{i=1}^n$ are Topological Spaces. For a map

$$f : X \to \prod_{i=1}^{n} Y_i : x \mapsto (f_1(x), f_2(x), \dots, f_n(x))$$

The Follwings Are Equivalent:

a) f is Continuous map.

b) For all $i = 1, 2, \dots, n$, $f_i : X \to Y_i$ is Continuous map.

Proof. $a) \implies b)$. Since π_i is Continuous map, $f_i = \pi_i \circ f$ be a Continuous map.
$b) \implies a)$. Let $B = U_1 \times U_2 \times \cdots \times U_n \in \mathcal{T}_P$, and $\pi_i : \prod_{i=1}^n Y_i \to Y_i$ be a Projection for each $i = 1, 2, \dots, n$. Then,

$$\begin{aligned} B = U_1 \times U_2 \times \cdots \times U_n = &(U_1 \times X_2 \times X_3 \times \cdots \times X_n) \\ &\cap (X_1 \times U_2 \times X_3 \times \cdots \times X_n) \\ &\vdots \\ &\cap (X_1 \times X_2 \times X_3 \times \cdots \times U_n) = \bigcap_{i=1}^{n} \pi_i^{-1}[U_i] \end{aligned}$$

Now,

$$f^{-1}[B] = f^{-1}\left[\bigcap_{i=1}^{n} \pi_i^{-1}[U_i]\right] = \bigcap_{i=1}^{n} f^{-1}\left[\pi_i^{-1}[U_i]\right] = \bigcap_{i=1}^{n} (\pi_i \circ f)^{-1}[U_i] = \underbrace{\overbrace{\bigcap_{i=1}^{n} f_i^{-1}[\underbrace{U_i}_{\text{open}}]}^{\text{finite intersection}}}_{f_i \text{ conti}} \in \mathcal{T}_X$$

□

Theorem 0.1.5. Let $X_1, \dots, X_n$ are Topological Space and $A_i \subset X_i$. Then, for a Product Space $\left(\prod_{i=1}^{n} X_i, \mathcal{T}_p\right)$,

1. $\overline{\prod_{i=1}^{n} A_i} = \prod_{i=1}^{n} \overline{A_i}$.
2. $\left(\prod_{i=1}^{n} A_i\right)^{\circ} = \prod_{i=1}^{n} A_i^{\circ}$.

Proof.
1.)

$$\begin{aligned} x \in \overline{\prod_{i=1}^{n} A_i} &\iff \forall U \in \mathcal{T}_p \text{ with } x \in U,\ U \cap \prod_{i=1}^{n} A_i = \prod_{i=1}^{n} \pi_i[U] \cap \prod_{i=1}^{n} A_i = \prod_{i=1}^{n} (\pi_i[U] \cap A_i) \neq \emptyset \\ &\iff \forall U \in \mathcal{T}_p \text{ with } x \in U,\ \forall i = 1, 2, \dots, n,\ \pi_i[U] \cap A_i \neq \emptyset \\ &\iff \forall i = 1, 2, \dots, n,\ \forall U_i \in \mathcal{T}_i \text{ with } \pi_i(x) \in U_i,\ U_i \cap A_i \neq \emptyset \\ &\iff x \in \prod_{i=1}^{n} \overline{A_i} \end{aligned}$$

2.)

$$\begin{aligned} x \in \left(\prod_{i=1}^{n} A_i\right)^{\circ} &\iff \exists U \in \mathcal{T}_p \text{ s.t } x \in U \subset \prod_{i=1}^{n} A_i \\ &\iff \exists U \in \mathcal{T}_p \text{ s.t } x \in \prod_{i=1}^{n} \pi_i[U] \subset \prod_{i=1}^{n} A_i \\ &\iff \exists U \in \mathcal{T}_p \text{ s.t } \forall i = 1, 2, \dots, n,\ \pi_i(x) \in \pi_i[U] \subset A_i \\ &\overset{(*)}{\iff} x \in \prod_{i=1}^{n} A_i^{\circ} \end{aligned}$$

However, the left direction of $(*)$ fails when the index set is infinite. $\square$